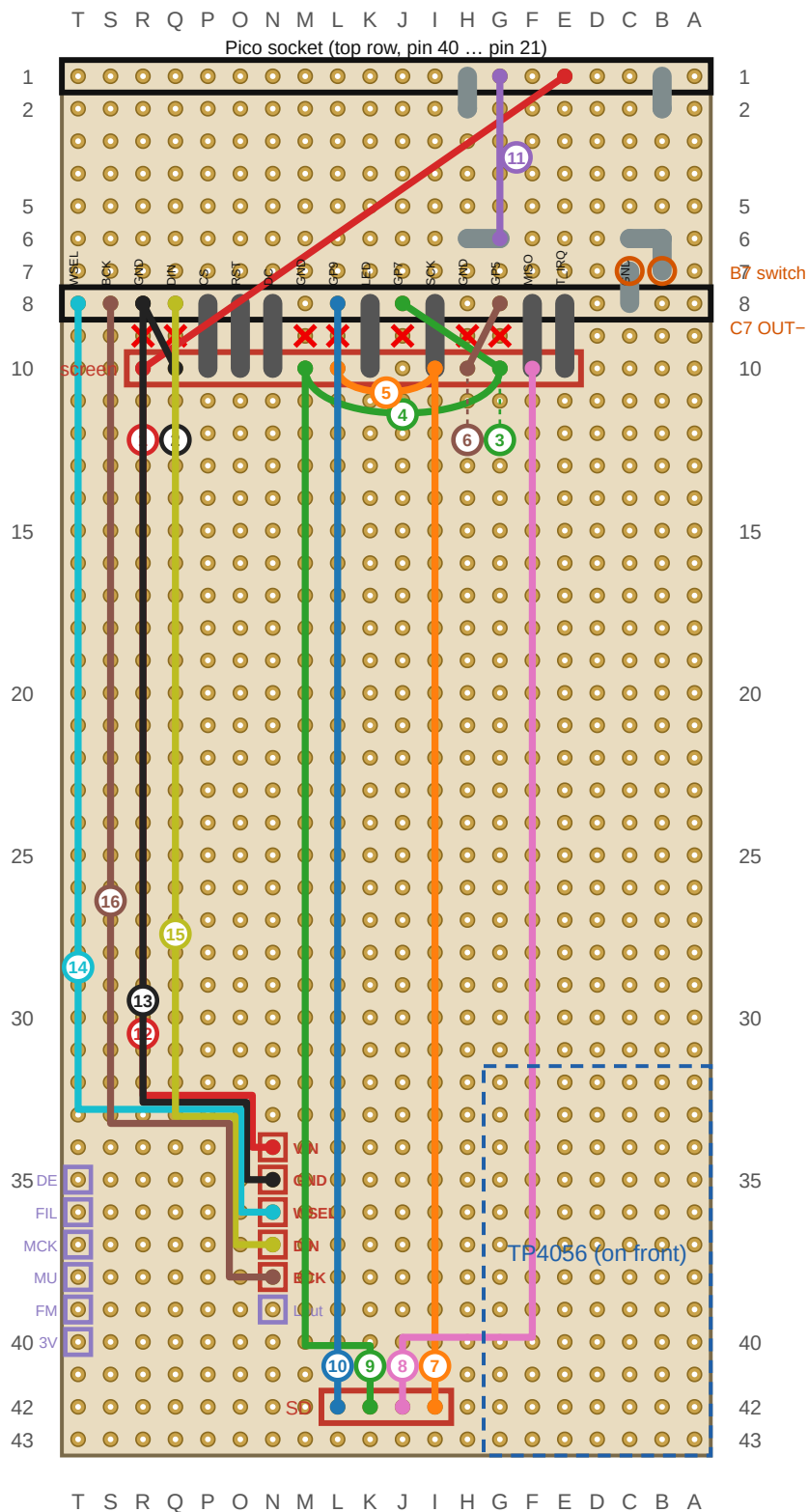


Pod perfboard: solder side



This is the board **turned over**: column A is on the right. All of this is on the solder side.

Dark grey bars (rows 8 to 10): 7 links. In columns E, F, I, K, N, O and P, the Pico pin sits directly above the screen pin it drives. Lay a cut-off resistor leg across holes 8, 9 and 10 and solder it. No wire needed.

Red X (row 9): never join these. In columns G, H, J, L, M, Q and R, the top and bottom are different signals (R8 is GND and R10 is 3.3 V, for example).

Light grey bars: solder bridges between neighbouring holes under the Pico: B1-B2, B6-B7, B6-C6, G6-H6, H1-H2, C7-C8.

Coloured lines: real wires, 30 AWG insulated wire-wrap wire. 16 in all: 11 for the screen and battery, 5 for the DAC.

#	From	To	Signal
1	E1	R10	3V3 → screen VCC
2	R8	Q10	GND → screen GND
3	J8	G10	GP7 → T_DIN (MOSI)
4	G10	M10	T_DIN → SDI (MOSI)
5	I10	L10	T_CLK → SCK
6	G8	H10	GP5 → T_CS
7	I10	I42	T_CLK → SD_SCK
8	F10	J42	T_DO → SD_MISO
9	M10	K42	SDI → SD_MOSI
10	L8	L42	GP9 → SD_CS
11	G6	G1	battery sense → GP28
12	R10	N34	3V3 → DAC VIN
13	R8	N35	GND → DAC GND
14	T8	N36	GP15 → DAC WSEL
15	Q8	N37	GP13 → DAC DIN
16	S8	N38	GP14 → DAC BCK

100 nF capacitor: across R2 (H2 and H6).

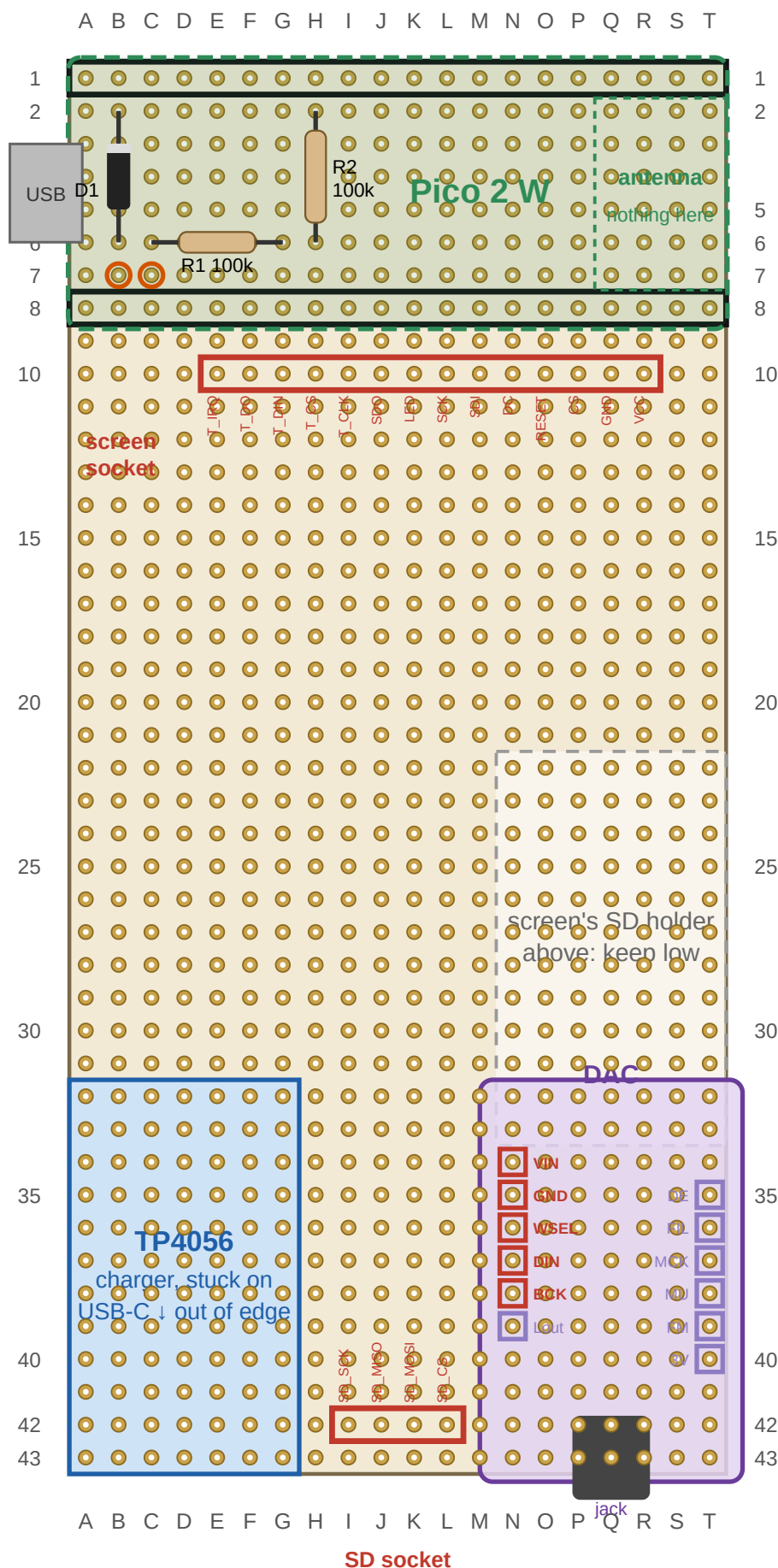
Hole-by-hole map

Every hole that gets solder, row by row. Tick each one as you do it. Any hole not listed stays empty.

	Hole	What's in it	Joins to
☐	B1	Pico pin 39 (VSYS)	solder bridge to B2
☐	E1	Pico pin 36 (3V3)	wire 1 starts here
☐	G1	Pico pin 34 (GP28)	wire 11 ends here
☐	H1	Pico pin 33 (AGND)	solder bridge to H2
☐	rest of row 1	Pico socket pins	socket solder joint only
☐	B2	D1, band end	bridge to B1
☐	H2	R2 + 100 nF	bridge to H1
☐	B6	D1, plain end	bridges to B7 and C6
☐	C6	R1	bridge to B6
☐	G6	R1	bridge to H6; wire 11 starts
☐	H6	R2 + 100 nF	bridge to G6
☐	B7	red wire from switch	bridge to B6
☐	C7	black wire from TP4056 OUT–	bridge to C8
☐	C8	Pico pin 3 (GND)	bridge to C7
☐	E8	Pico pin 5 (GP3)	LINK to E10
☐	F8	Pico pin 6 (GP4)	LINK to F10
☐	G8	Pico pin 7 (GP5)	wire 6 starts
☐	I8	Pico pin 9 (GP6)	LINK to I10
☐	J8	Pico pin 10 (GP7)	wire 3 starts
☐	K8	Pico pin 11 (GP8)	LINK to K10
☐	L8	Pico pin 12 (GP9)	wire 10 starts
☐	N8	Pico pin 14 (GP10)	LINK to N10
☐	O8	Pico pin 15 (GP11)	LINK to O10
☐	P8	Pico pin 16 (GP12)	LINK to P10
☐	Q8	Pico pin 17 (GP13)	wire 15 starts
☐	R8	Pico pin 18 (GND)	wires 2 and 13 start
☐	S8	Pico pin 19 (GP14)	wire 16 starts
☐	T8	Pico pin 20 (GP15)	wire 14 starts
☐	A8, B8, D8, H8, M8	Pico socket pins	socket solder joint only
☐	E9	link only	part of the E8–E10 link
☐	F9	link only	part of the F8–F10 link

	Hole	What's in it	Joins to
☐	I9	link only	part of the I8–I10 link
☐	K9	link only	part of the K8–K10 link
☐	N9	link only	part of the N8–N10 link
☐	O9	link only	part of the O8–O10 link
☐	P9	link only	part of the P8–P10 link
☐	other row 9	empty	never solder G9, H9, J9, L9, M9, Q9, R9
☐	R10	screen pin 1 (VCC)	wire 1 ends; wire 12 starts
☐	Q10	screen pin 2 (GND)	wire 2 ends
☐	P10	screen pin 3 (CS)	LINK from P8
☐	O10	screen pin 4 (RESET)	LINK from O8
☐	N10	screen pin 5 (DC)	LINK from N8
☐	M10	screen pin 6 (SDI)	wires 4 and 9
☐	L10	screen pin 7 (SCK)	wire 5 ends
☐	K10	screen pin 8 (LED)	LINK from K8
☐	J10	screen pin 9 (SDO)	nothing (SDO stays unconnected)
☐	I10	screen pin 10 (T_CLK)	LINK from I8; wires 5 and 7
☐	H10	screen pin 11 (T_CS)	wire 6 ends
☐	G10	screen pin 12 (T_DIN)	wires 3 and 4
☐	F10	screen pin 13 (T_DO)	LINK from F8; wire 8 starts
☐	E10	screen pin 14 (T_IRQ)	LINK from E8
☐	L42	SD pin 1 (SD_CS)	wire 10 ends
☐	K42	SD pin 2 (SD_MOSI)	wire 9 ends
☐	J42	SD pin 3 (SD_MISO)	wire 8 ends
☐	I42	SD pin 4 (SD_SCK)	wire 7 ends
☐	N34	DAC pin VIN	wire 12 ends
☐	N35	DAC pin GND	wire 13 ends
☐	N36	DAC pin WSEL	wire 14 ends
☐	N37	DAC pin DIN	wire 15 ends
☐	N38	DAC pin BCK	wire 16 ends
☐	N39	DAC pin Lout	nothing
☐	T35–T40	DAC pins DE, FIL, MCK, MU, FM, 3V	nothing : solder joint only

Front view: where everything sits



Holes are named by column letter (A–T, left to right on the front) and row number (1–43, top to bottom). Example: **B7**.

Orange rings: B7 and C7, where the switch wire and the TP4056 OUT- wire come in.

Black boxes: two 1×20 sockets for the Pico, rows 1 and 8, all 20 columns. USB end on the left.

Red boxes: screen sockets. 1×14 in row 10, columns E–R. 1×4 in row 42, columns I–L.

Under the Pico (it sits 8.5 mm up, on its sockets): D1 from B2 (band end) to B6, R1 from C6 to G6, R2 from H2 to H6. Lay them flat.

Antenna end (right): no wires or parts on the front under it.

Bottom left: TP4056, stuck on with foam tape, USB-C poking 1 mm past the bottom edge.

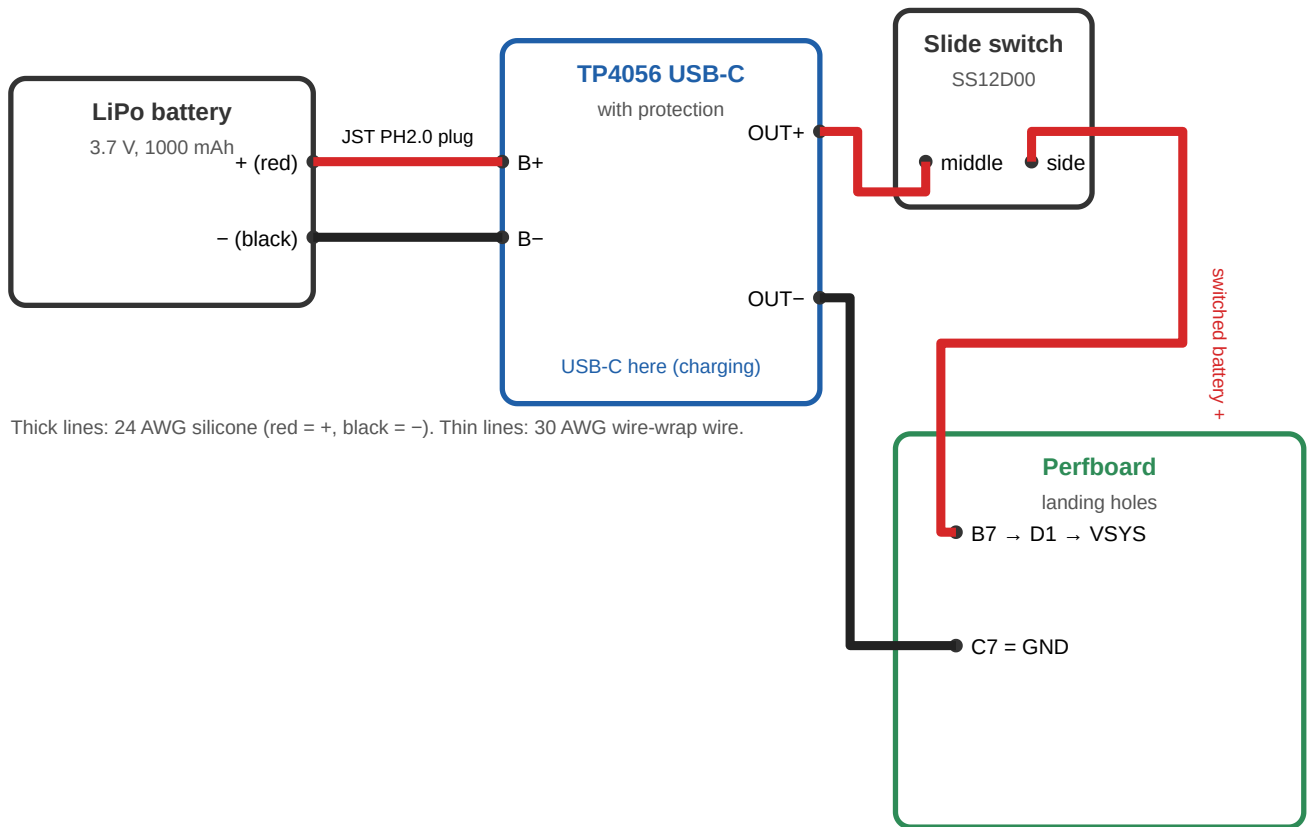
Bottom right: DAC, jack down at the bottom edge. Its wire row goes in column N (N34–N39), the other row in column T (T35–T40). It overhangs the right edge by about 1 mm.

Grey dashed area: the screen's SD-card holder sits above it. Keep it clear.

Everything under the screen must stay under **10 mm** tall.

Cutting the board: score along a line of holes on both sides, 6–8 times with the box cutter against the ruler, then snap it over a table edge and sand. You need 20 holes across and 43 down.

Off-board wires: battery, charger, switch



Build order

1. Cut the board. Solder the two Pico sockets (plug the Pico in first so they line up, then solder).
2. Solder the 1×14 screen socket in row 10. For the 1×4 SD socket: plug it onto the screen's SD pins, plug the screen into the 1×14 socket, push the 1×4 through row 42, then solder. This lines it up exactly.
3. D1, R1, R2, the capacitor and the six solder bridges.
4. The 7 links, then wires 1–10. **Test:** Pico in, screen on, USB from the laptop only (no battery). Home screen, touch and SD card should work.
5. DAC: pins through N34–N39 and T35–T40, solder, then wires 12–16. **Check first:** N35 must beep to the jack's outer metal tab (that pin is GND). **Test:** play music from the SD card on headphones.
6. Wire 11, then the TP4056, switch and battery wires. **Before plugging in the battery:** with the multimeter on continuity, check B7 to C7 does **not** beep, and check red really goes to B+.
7. Plug in the battery, switch on. Kapton tape over the back, then stick the battery on with foam tape.

Shopping list

Item	Qty	Notes
Perfboard, double-sided, plated holes , 2.54 mm, 9 × 15 cm	1	green FR-4, not brown paper board. Cut to 20 × 43 holes.
Female header strip 1 × 40, 2.54 mm (8.5 mm tall)	3	Pico 2 × 20, screen 1 × 14, SD 1 × 4. Each cut wastes one pin.
TP4056 charger module, USB-C, with protection	1–2	must have 6 pads: IN+, IN–, B+, B–, OUT+, OUT–
LiPo 3.7 V, 1000 mAh, size 503450 (5 × 34 × 50 mm)	1	with protection board and JST PH2.0 plug
JST PH2.0 2-pin socket with wires	1	matches the battery's plug
Slide switch SS12D00 (3 pins)	2	power switch, fitted in the case wall
Schottky diode 1N5819	2	D1
Resistor 100 kΩ, ¼ W	3	R1, R2
Ceramic capacitor 100 nF ("104")	2	soldered across R2
Wire-wrap wire, 30 AWG (Kynar), 2–3 colours	1 roll each	all signal wires on the back
Silicone wire, 24 AWG stranded, red + black	1 m each	battery, charger, switch
Solder 0.6–0.8 mm with flux, flux pen, desolder wick		
Kapton tape, double-sided foam tape, 2 mm heat-shrink		insulate the back before the battery goes on

Tools: temperature-controlled iron (330–350 °C), flush cutters, a stripper that handles 30 AWG, **multimeter**, box cutter and steel ruler.

Which wire where

- **Signals (the 11 numbered wires):** 30 AWG wire-wrap wire, single core. Thin, stays where you bend it, and its insulation doesn't melt back when you solder next to it.
- **Battery, charger, switch:** 24 AWG stranded silicone. Flexible, carries over 1 A. Red for +, black for –.
- **Neighbouring holes** (marked grey on the back view): join with a blob of solder, or a scrap of cut-off resistor lead.
- No Dupont jumper wires in the final build: they're tall and work loose.

Battery and charging

- The battery only connects to the **TP4056** (B+ / B–), never straight to the Pico.
- TP4056 **OUT+** → **power switch** → **D1** → **Pico VSYS**. D1 stops the Pico's USB from pushing 5 V into the battery, so you can plug in the Pico's micro-USB to flash firmware at any time.
- Charge through the TP4056's USB-C. Red LED = charging, blue/green = full. It charges at 1 A, which is fine for a 1000 mAh cell. For a proper full charge, switch the Pod off while it charges.
- R1 and R2 halve the battery voltage so GP28 can read it. The firmware needs a small update to show the percentage from this; until then the battery icon stays hidden and everything else works.
- **Check the plug polarity** before connecting: cheap PH2.0 batteries are sometimes wired backwards. Red must reach B+. Never cut both battery wires at once.